

Samuel Schiavone

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PROFILE

I am a researcher, programmer, and mathematician who combines mathematical abstraction with computational tools in order to discover, analyze, and elucidate. My expertise lies in translating theoretical concepts into practical solutions.

SKILLS

Programming Languages: Python, PostgreSQL, SageMath, Magma, R

Programming Tools: Git, Flask, Jinja, GNU Parallel, Scikit-learn, Pandas

Languages: English (native), French (fluent), Spanish (intermediate)

SELECTED PROJECTS

Development for the L-functions and modular forms database (LMFDB)

May 2018 – Present

Full Stack Web Developer

Remote

- Collaborated on a joint open-source project to create an accessible repository of data in algebra and number theory, used by over 30,000 researchers and educators in math
- Co-created two sections of an online database: added data to the backend using Python and PostgreSQL; designed webpages to display the data using Flask, Jinja, and HTML
- Supervised four undergraduate students in the creation of a Magma and SageMath package to compute additional data
- Connected to other sections of the LMFDB by identifying and linking to related objects using scripts written in Python and SageMath

Computations in Inverse Galois Theory

Jan 2022 – Nov 2024

Academic Researcher

Cambridge, Massachusetts

- Collaborated on a team of 6 researchers to solve an open mathematical problem in computational number theory
- Combined theoretical expertise with practical programming skills in Python, PostgreSQL, SageMath, and Magma to produce a research article and two repositories of code, one a general Magma package and one tailored to a specific example
- Presented results in two invited talks to audiences of 30-50 researchers at VaNTAGe and Simons AGNTC annual meeting
- Synthesized detailed understanding of theoretical results in arithmetic geometry with technical programming expertise to give an explicit solution to an open mathematical problem

EXPERIENCE

Research Scientist

Sep 2019 – Aug 2025

Massachusetts Institute of Technology

Cambridge, Massachusetts

- Conducted research in computational arithmetic geometry
- Devised and implemented algorithms to compute mathematical objects, wrote mathematical proofs, and chronicled my research in academic articles
- Aided in the development and maintenance of the L-functions and Modular Forms Database (<https://www.lmfdb.org/>), working both with data in PostgreSQL and Python on the backend and in Flask, Jinja, and HTML on the frontend
- Instructed undergraduate students in courses on linear algebra, algebraic geometry, mathematical communication, and Belyi maps

EDUCATION

Dartmouth College

Ph.D. in Mathematics

Hanover, New Hampshire

Sep 2019

The University of Vermont

M.S. in Mathematics

Burlington, Vermont

Dec 2013

Amherst College

B.A. in Mathematics, magna cum laude

Amherst, Massachusetts

May 2010